KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arXiv:2504.03553 · ALFWorld Experiments · 2025

For ML researchers and graduate students in LLM agents & planning systems

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

1 Trajectory Imitation

ReAct-style direct behavior cloning

- No self-correction signal
- Pattern collapse on novelty
- Blind to unknown states

2 Trial & Error Refinement

Reflexion-style feedback loops

- Expensive environment steps
- May reinforce bad habits
- No external knowledge integration

3 Knowledge-Augmented

KnowAgent / WKM full KB injection

- 100% knowledge at every step
- Redundant queries for easy cases
- Hurts efficiency & can misguide

💡 Humans use metacognition to decide: act directly, reflect, or seek help — KnowSelf brings this to LLM agents.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking

KnowSelf identifies the agent's cognitive state at each decision step

↑ YES

⚡ Fast Thinking

Direct correct action
with minimal deliberation
→ Act immediately

Token: <Fast>

YES →

Can agent produce correct action now?

NO →

Can self-reflection correct it?

↓ NO

🔁 Slow Thinking

Token: <Reflect>

📖 Knowledgeable Thinking

Cannot reach correct
answer even after reflection
→ Retrieve external knowledge

Token: <Know>

Special tokens (<Fast>, <Reflect>, <Know>) are inserted during training via heuristic labeling on self-explored trajectories. At inference, generated tokens trigger the corresponding action or retrieval.

KnowSelf Framework: Data → Training → Inference

A three-stage pipeline for agentic self-aware knowledge regulation

1 Self-Aware Data Construction

Agent self-explores environments

Heuristic labels each step:

✓ Gold match → Fast

✗ Fix by refl → Slow

✗ Needs help → Knowledge

Special tokens mark each state

2 Two-Stage Training

Stage A: SFT

Supervised fine-tuning on

token-marked trajectories

Stage B: RPO

Reward-based Policy Opt.

trains token generation

3 Self-Aware Inference

Agent generates special token

<Fast> → act directly

<Reflect> → internal reflection

<Know> → query external KB

Minimal knowledge footprint

Figure: KnowSelf pipeline (data construction → SFT+RPO training → self-aware inference). Token-based triggering enables selective knowledge retrieval at inference time.

KnowSelf Correctly Heats a Mug and Places Objects Where Baselines Fail

Qualitative comparisons on ALFWorld tasks (figure_1.jpg reference)

Task 1: 'Put a hot mug in cabinet'

✅ KnowSelf (Llama-8B) — SUCCESS ✓

Uses <Reflect> token to self-correct:

→ Realizes mug must be heated first

→ Uses open microwave to heat mug

→ Completes task correctly

❌ OpenAI-O1 — FAILURE ✗

Attempts to take egg from microwave

after opening it — wrong object

No self-correction triggered

Self-awareness via special tokens prevents errors that

Task 2: 'Put a saltshaker in drawer'

✅ KnowSelf (Llama-8B) — SUCCESS ✓

Uses <Know> token to retrieve knowledge:

→ Places object directly in drawer

→ Does not remove unrelated items

→ Applies correct object placement rule

❌ DeepSeek-R1 — FAILURE ✗

Reflects: 'drawer is open'

Tries to close it first (fails)

No knowledge retrieval triggered

Self-awareness via special tokens prevents errors that

These examples illustrate how selective reflection and knowledge retrieval — triggered only when needed — outperform models that either over-reflect or under-retrieve.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B — Using Only 15% Knowledge

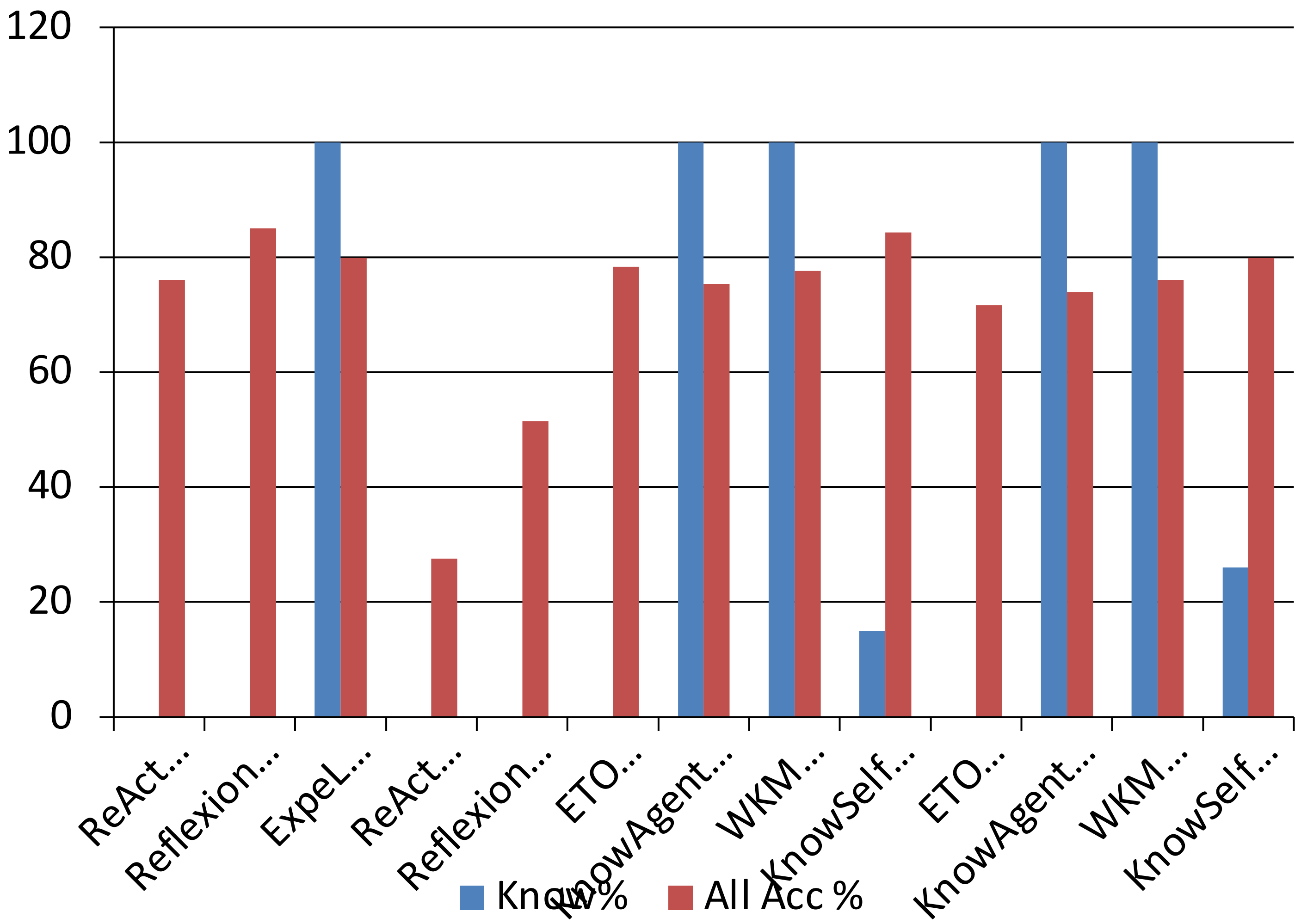
Success rates (%) across six ALFWorld task types (Know% = external knowledge usage)

GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61

KnowSelf (bold orange) achieves highest All scores using far less knowledge (15%/26%) than 100% knowledge methods. Green = 0% knowledge; Red = 100% knowledge.

Minimal Knowledge, Maximum Impact: Efficiency Advantage Over Full-Knowledge Methods

KnowSelf uses only ~15–26% knowledge queries vs. 100% for competing methods



Efficiency Summary

15% — **KnowSelf (Llama-8B) Knowledge Usage**

vs. 100% for KnowAgent/WKM/ExpeL

84.33 % — **Overall Accuracy ALFWorld All Tasks**

Best overall — surpasses GPT-4o Reflexion

79.85 % — **KnowSelf (Gemma-2B) Overall Accuracy**

Matches GPT-4o ExpeL at 1/50th model scale

26% — **KnowSelf (Gemma-2B) Knowledge Usage**

vs. 100% for competing Gemma-2B methods

Key insight: indiscriminate 100% knowledge injection often underperforms selective ~15–26% usage. KnowSelf proves that knowing when NOT to query is as important as querying.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

Five findings that define the KnowSelf contribution

01

Indiscriminate knowledge injection is wasteful and often counterproductive

Methods using 100% knowledge queries (ExpeL, KnowAgent, WKM) frequently underperform KnowSelf's selective ~15–26% approach. More knowledge is not always better.

02

Agents can learn situational self-awareness via lightweight heuristic token labeling

No expensive human annotation or massive external feedback loops required. Self-exploration + simple heuristics → special tokens → self-aware behavior.

03

KnowSelf with Llama-8B (84.33%) outperforms strong GPT-4o baselines

Exceeds GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) while using only 15% of the external knowledge. Demonstrates the power of selectivity.

04

Only 15–26% knowledge usage needed — not 100%

KnowSelf's token-triggered retrieval ensures knowledge is consulted only when reflection cannot produce a correct answer. Efficiency and accuracy co-exist.

05

The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%

KnowSelf on Gemma-2B matches GPT-4o ExpeL, exceeding all other Gemma-2B baselines, showing that self-awareness is a model-scale-agnostic architectural property.